

A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Mineral resources (the lithosphere) - Set 1

Mark scheme

AQA spec 3.2.3; The physical environment; textbook pp. 130-147

Marking guidance

- The number in brackets after each marking point is the mark it earns; the note above the points says how many are needed.
- Where a question asks for a fixed number of points (eg *two reasons*), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	<i>1 mark for each point.</i> • Igneous intrusions (batholiths) of magma cool within the crust, heating groundwater to produce superheated, pressurised water (1) • The hot water dissolves minerals from the surrounding rock and the mineral-rich solutions travel along fissures / cracks away from the batholith (1) • As the solutions cool the dissolved minerals crystallise in order of solubility (least soluble first) - fractional crystallisation (1) • So different metals are separated and concentrated in veins, eg tin, copper, lead, silver, gold, arsenic (SW England batholith) (1)	4
01.2	<i>1 mark for each point.</i> • Dissolved iron compounds in ancient seas were oxidised by the oxygen released by early photosynthetic organisms (2.5-1.8 billion years ago) (1) • Iron oxides are insoluble so they were deposited as sediments (haematite, magnetite) (1)	2
01.3	<i>1 mark for each point.</i> • Flowing water carries solid particles; the ability to carry them depends on the water velocity and the density of the solids (1) • As velocity falls, dense minerals (gold, tin ore) are deposited first while lighter material is carried further, concentrating the heavy minerals in particular places (1)	2
01.4	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> • Limestone / chalk from the shells of marine organisms (1) • Coal from terrestrial vegetation (1) • Crude oil and natural gas from marine organisms (plankton) (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 132-134; AQA spec 3.2.3.2.	

Question	Marking guidance	Marks
02.1	<i>1 mark.</i> As the (linear) purity / grade of a mineral deposit decreases, the amount of the mineral present increases exponentially / logarithmically (1)	1
02.2	<i>1 mark for the correct answer.</i> $2400 / 40 = 60$ times (1)	1
02.3	<i>1 mark for each point.</i> - Reserves: the amount that can be exploited now, economically, with existing technology (deposits above the cut-off ore grade) (1) - Resource: all the material that could be exploited technically and economically now or in the future (with realistic price rises or new technology), including deep, low-grade or protected deposits (1) - The resource is finite but the reserves change: they rise if prices rise or technology improves, and fall if prices fall (1)	3
02.4	<i>1 mark for each point.</i> - Lowering the cut-off ore grade includes deposits of lower purity in the reserves (1) - Because the amount of material rises exponentially as purity falls, a small drop in COOG adds a very large mass (eg 0.5-1.0% has 4x the copper of 1.0-2.0%) (1)	2
02.5	<i>1 mark for each point.</i> - The total quantity (stock / resource base) is far greater than could ever be used (1) - But most is at purities too low or in locations too inaccessible for current economic exploitation (1) - The challenge is developing technologies (low-energy extraction such as bioleaching, phytomining; better exploration) that make low-grade deposits exploitable; energy needs rise rapidly as ore grade falls (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 135-136; AQA spec 3.2.3.3.	
03.1	<i>1 mark for each point.</i> - Gravimetry: gravimeters detect variations in gravity caused by differences in rock density / mass (metal ores usually denser than sedimentary rock) (1) - Magnetometry (1) - Seismic survey: sound / shock waves from controlled explosions or a seismic vibrator; echoes recorded by geophones give the depth, density and shape of rock strata (1) - Resistivity (1)	4
03.2	<i>1 mark for each point.</i> - Trial drilling is the most expensive technique per site (1) - Remote sensing (satellite, aircraft, ground surveys) covers large areas at low cost per unit area and identifies promising sites (1) - Drilling is then targeted where deposits are likely; it is the only method that produces actual rock samples for chemical analysis to confirm composition and purity (proven reserves) (1)	3
03.3	<i>1 mark for each point.</i> - Different minerals emit / reflect infrared at different wavelengths (1) - Multispectral sensors detect these wavelengths over the whole surface (eg Landsat 8 every 16 days) giving information on surface mineral composition (1)	2

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03.4	<i>1 mark.</i> Results are available immediately in the field, saving time waiting for laboratory analysis (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 137; AQA spec 3.2.3.4.	
04.1	<i>1 mark for each point.</i> - The lowest ore purity that can be mined economically with existing technology (1) - Deposit X (0.9% is above 0.4%; Y at 0.35% is part of the resource, not the reserves) (1)	2
04.2	<i>1 mark for each point.</i> - X has higher grade so less rock mined, less spoil, less energy and pollution per tonne of copper (1) - But X is deep: costs rise more than proportionally with depth (sides cannot be vertical so much more rock removed; more water inflow and pumping) (1) - X has hard overburden requiring blasting (cost, noise, dust) and high rainfall increasing drainage costs (1) - X is far from transport infrastructure so transport costs are high (1) - Y is shallow with loose overburden (cheap to remove but landslide risk so gentler slopes / larger land take), near a railway, low rainfall (1) - Y's grade is below the COOG so it is not viable now but would become viable if copper prices rose or extraction technology improved (eg bioleaching); both sulfides risk acid mine drainage (1)	6
04.3	<i>Any 2 from the points below - 1 mark each, maximum 2.</i> - Higher market value / price of the metal (1) - Improved extraction technology (1) - Lower energy costs (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 138-139; AQA spec 3.2.3.5.	
05.1	<i>1 mark for each point.</i> - Minerals can only be exploited where they were deposited (no choice of location), unlike housing, roads or ports (1) - Land take is larger than the mine void: buildings, access roads, overburden and spoil heaps, buffer zones; open-cast mines take far more land than deep mines (1)	2
05.2	<i>1 mark for each point.</i> - Dust: water sprays on roads / blasting areas make particles heavier and clump so they settle (1) - Noise: baffle mounds / embankments absorb and deflect; blasting at set, predictable daytime hours; quieter machinery (1) - Turbid water: sedimentation lagoons where water stands until solids settle before discharge (1) - Leachate: collect drainage water and pass through crushed limestone to neutralise acid and immobilise toxic metals (1) - Amenity: landscaping and tree planting to screen the mine; restoration plans agreed as a condition of planning permission (1)	5

Question	Marking guidance	Marks
05.3	<i>1 mark for each point.</i> - Landscaping to reduce gradients, compaction, drainage to prevent waterlogging, adding soil / nutrients and planting vegetation whose roots bind the surface (1) - At Aberfan a waterlogged coal-mine spoil heap became unstable and slid into the village (1) - Killing 144 people, mainly children in the primary school (1)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 139-142; AQA spec 3.2.3.6.	
06.1	<i>1 mark for each point.</i> - The void fills with water naturally; wetlands are uncommon / valuable habitats in most regions (1) - The new habitat may have greater wildlife value than the farmland lost; low cost compared with filling and levelling (1)	2
06.2	<i>1 mark for each point.</i> - Housing: ground must be stable (no subsidence), no toxic metal wastes (otherwise industrial use is safer) (1) - Farmland: landscape not too steep or uneven, no toxic materials, sufficiently fertile soil (topsoil replaced) (1)	2
06.3	<i>1 mark.</i> Species that cannot be caught / found, or whose habitat requirements are unknown or hard to recreate, are lost; transplanted plants often fail (1)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 140, 142; AQA spec 3.2.3.6.	
07.1	<i>1 mark for each point.</i> - Acidophilic bacteria (eg Acidithiobacillus) oxidise sulfide ores, producing sulfuric acid which dissolves the metal ions (1) - Leachate (rainwater / recirculated solution) percolating through the spoil heap is collected; recirculation increases the metal concentration (1) - Copper is recovered from solution by electrolysis or by displacement with scrap iron (iron is more reactive so copper is deposited as solid metal) (1) - Smelting needs chemical reduction at high temperatures; energy needed rises rapidly as ore grade falls, so low-grade ores are uneconomic (1) - Bioleaching works at low temperatures with low energy inputs, so it can exploit low-grade deposits and wastes (increasing reserves) - mimics natural low-temperature processes (1)	5
07.2	<i>1 mark for each point.</i> - Plants (hyperaccumulators) absorb metal ions from soil / water and concentrate them in their leaves (1) - The vegetation is harvested and incinerated (1) - The concentrated metal in the ash is dissolved in acid and separated by electrolysis; also decontaminates polluted sites (1)	3
07.3	<i>1 mark for each point.</i> - Uranium ions dissolved in seawater adsorb onto the surface of synthetic or natural polymers (lignin from wood, chitin from shrimp shells) (1) - The polymers are collected and the uranium recovered, avoiding mining / high-energy extraction (1)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 143-145; AQA spec 3.2.3.7.	

Question	Marking guidance	Marks
08.1	<i>1 mark for each point.</i> • Metal-rich lumps 5-10 cm across on the deep seabed (4000-5000 m), containing about 30% manganese plus nickel, copper, cobalt, iron, titanium (1) • Exploitation would be very expensive (remotely operated collectors, pumping to the surface) / no viable technology yet (1) • International agreement on ownership of the seabed and its resources is needed (1)	3
08.2	<i>1 mark for each point.</i> • Collectors disturb the seabed and kill benthic organisms living on it (1) • Separating nodules from sediment creates plumes that increase turbidity with unknown consequences (1) • Re-deposition of sediment covers and kills filter feeders and other benthic organisms; slow recovery in the deep sea (1)	3
08.3	<i>1 mark for each point.</i> • Used in small amounts in many essential technologies (smartphones, hybrid vehicles, wind turbines, fibre optics, LEDs) (1) • Exploitable land deposits are limited and mainly in China; chemical separation is complex and produces large amounts of toxic waste (1) • Bacterial adsorption: dissolved rare earth ions adsorb onto the cell surfaces of bacteria (<i>Bacillus subtilis</i>) and are washed off at specific pHs to separate them (1) • Deep sea sediments contain relatively high concentrations but no viable extraction method yet (1)	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 145-146; AQA spec 3.2.3.7.	